

Questions with Answer Keys

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Q1 - Single Correct

The value of $\lim_{n \rightarrow \infty} \frac{1^2 \cdot n + 2^2 \cdot (n-1) + 3^2 \cdot (n-2) + \dots + n^2 \cdot 1}{1^3 + 2^3 + 3^3 + \dots + n^3}$

- (1) $\frac{1}{3}$
- (2) $\frac{2}{3}$
- (3) $\frac{1}{2}$
- (4) $\frac{1}{6}$

Q2 - Single Correct

The value of $\lim_{x \rightarrow \infty} \frac{\cot^{-1}(x^{-a} \log_a x)}{\sec^{-1}(a^x \log_x a)}, (a > 1)$ is

- (1) 1
- (2) 0
- (3) $\frac{\pi}{2}$
- (4) Does not exist

Q3 - Single Correct

Given, $I_1 = \lim_{x \rightarrow \frac{\pi}{4}} \cos^{-1} \left[\sec \left(x - \frac{\pi}{4} \right) \right]; I_2 = \lim_{x \rightarrow \frac{\pi}{4}} \sin^{-1} \left[\operatorname{cosec} \left(x + \frac{\pi}{4} \right) \right];$
 $I_3 = \lim_{x \rightarrow \frac{\pi}{4}} \tan^{-1} \left[\cot \left(x + \frac{\pi}{4} \right) \right]; I_4 = \lim_{x \rightarrow \frac{\pi}{4}} \cot^{-1} \left[\tan \left(x - \frac{\pi}{4} \right) \right]$, where $[\cdot]$ denotes greatest integer function, then which of the following limits exist?

- (1) I_1 and I_2
- (2) I_1 and I_3
- (3) I_1 and I_4
- (4) All of these

Q4 - Single Correct

The value of $\lim_{x \rightarrow 0} \frac{\tan(\{x\}-1) \sin\{x\}}{\{x\}(\{x\}-1)}$ (where, $\{x\}$ denotes the fractional part of x), is

- (1) 1

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(2) $\tan 1$ (3) $\sin 1$

(4) non-existent

Q5 - Single Correct

If $P_n = \prod_{k=1}^n \left(1 + \frac{2}{k^2+3k}\right)$, then $\lim_{n \rightarrow \infty} P_n$ is equal to

(1) 1

(2) 3

(3) 5

(4) 7

Q6 - Single Correct

For $n \in \mathbb{N}$, let $a_n = \sum_{k=1}^n 2k$ and $b_n = \sum_{k=1}^n (2k-1)$. Then, $\lim_{n \rightarrow \infty} (\sqrt{a_n} - \sqrt{b_n})$ is equal to

(1) 1

(2) $\frac{1}{2}$

(3) 0

(4) 2

Q7 - Single Correct

The value of $\lim_{x \rightarrow 0} (\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x])$ (where, $[\cdot]$ denotes greatest integer function), is

(1) π (2) $\frac{\pi}{2}$ (3) $\frac{3\pi}{2}$

(4) Does not exist

Q8 - Single Correct

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Let $I = \lim_{x \rightarrow 2} \frac{[x]}{\{x\}}$ and $m = \lim_{x \rightarrow 2} \frac{\{x\}}{[x]}$, where $[\cdot]$ and $\{\cdot\}$ denotes greatest integer function and fractional part of x , then

- (1) I exists but m does not
- (2) m exists but I does not
- (3) both I and m exist
- (4) neither I nor m exists

Q9 - Single Correct

The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r^2}{n^3 + n^2 + r}$ is

- (1) $\frac{1}{3}$
- (2) $\frac{1}{2}$
- (3) $\frac{2}{3}$
- (4) 1

Q10 - Single Correct

Let $\lim_{x \rightarrow 0} \frac{[x]^2}{x} = I$ and $\lim_{x \rightarrow 0} \frac{[x^2]}{x^2} = m$ (where, $[\cdot]$ denotes greatest Integer function), then

- (1) I exists but m does not exist
- (2) m exists but I does not exist
- (3) I and m both exist
- (4) neither I nor m exists

Q11 - Single Correct

Given, $f(x) = \frac{e^x - \cos 2x - x}{x^2}$ for $x \in \mathbb{R} - \{0\}$ and $g(x) = \begin{cases} f(\{x\}), & \text{for } n < x < n + \frac{1}{2} \\ f(1 - \{x\}), & \text{for } n + \frac{1}{2} \leq x < n + 1, n \in \mathbb{I}, \\ 5/2, & \text{otherwise} \end{cases}$

where $\{x\}$ denotes fractional part of x , then $g(x)$ is

- (1) discontinuous at all integral values of x only

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(2) continuous everywhere except for $x = 0$ (3) discontinuous at $x = n + 1/2; n \in I$ and at some $x \in I$

(4) continuous everywhere

Q12 - Single Correct

Let $f(x) = \frac{2 \cos x - \sin 2x}{(\pi - 2x)^2}$, $g(x) = \frac{e^{-\cos x} - 1}{8x - 4\pi}$ and $h(x) = \begin{cases} f(x), & \text{for } x < \pi/2 \\ g(x), & \text{for } x > \pi/2 \end{cases}$, then which of the following holds?

(1) h is continuous at $x = \pi/2$ (2) h has an irremovable discontinuity at $x = \pi/2$ (3) h has a removable discontinuity at $x = \pi/2$ (4) $f\left(\frac{\pi^+}{2}\right) = g\left(\frac{\pi^-}{2}\right)$

Q13 - Single Correct

Consider the function $f(x) = \begin{cases} x\{x\} + 1, & 0 \leq x < 1 \\ 2 - \{x\}, & 1 \leq x \leq 2 \end{cases}$, where $\{x\}$ denotes the fractional part of x . Which one of the following statement is incorrect?

(1) $\lim_{x \rightarrow 1} f(x)$ exists(2) $f(0) \neq f(2)$ (3) $f(x)$ is continuous in $[0, 2]$

(4) None of these

Q14 - Single Correct

If $f(x + y) = f(x) + f(y) + |x|y + xy^2, \forall x, y \in R$ and $f'(0) = 0$, then

(1) f need not be differentiable at every non-zero x (2) f is differentiable for all x (3) f is twice differentiable at $x = 0$

(4) None of the above

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Q15 - Single Correct

Let $f(x) = \max \{|x^2 - 2||x|, |x|\}$ and $g(x) = \min \{|x^2 - 2||x|, |x|\}$, then

- (1) both $f(x)$ and $g(x)$ are non-differentiable at 5 points
- (2) $f(x)$ is not differentiable at 5 points and $g(x)$ is not differentiable at 7 points
- (3) number of points of non-differentiability for $f(x)$ and $g(x)$ are 7 and 5 respectively
- (4) both $f(x)$ and $g(x)$ are non-differentiable at 3 and 5 points respectively

Q16 - Multiple Correct

Let $f(x) = \min(x^3, x^2)$ and $g(x) = [x]^2 + \sqrt{\{x\}^2}$, where $[\cdot]$ and $\{\cdot\}$ denote greatest integer and fractional part of x . Then, which of the following holds?

- (1) f is continuous for all x
- (2) g is discontinuous for all $x \in I$
- (3) f is differentiable for all $x \in (1, \infty)$
- (4) g is not differentiable for all $x \in I$

Q17 - Multiple Correct

The function, $f(x) = [|x|] - |[x]|$, where $[x]$ denotes greatest integer function

- (1) is continuous for all positive integers
- (2) is discontinuous for all non-positive integers
- (3) has finite number of elements in its range
- (4) is such that its graph does not lie above the X -axis

Q18 - Multiple Correct

Which of the following statement(s) is (are) correct?

- (1) If $f(x)$ is differentiable, $\forall x \in [0, 1]$, then $f'(x)$ must be bounded in $[0, 1]$
- (2) There exists a bijective function $f : [0, 1] \rightarrow [0, 1]$ which is not continuous
- (3) Let $f : [-1, 2] \rightarrow R$ be defined as $f(x) = x^2 - \cos \pi x + 4$, then $f(c) = 2\pi$ for some $c \in (-1, 2)$

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(4) If $f(x) = 3 + x + e^x$, then $(f^{-1})'(4)$ is equal to $(1/2)$

Q19 - Multiple Correct

Let $f(x)$ be twice differentiable function such that $f''(x) < 0$ in $[0, 2]$, then

(1) $f(0) + f(2) = 2f(c), 0 < c < 2$

(2) $f(0) + f(2) = 2f(1)$

(3) $f(0) + f(2) > 2f(1)$

(4) $f(0) + f(2) < 2f(1)$

Q20 - Multiple Correct

Let $f(x) = \begin{cases} |x|^p \sin \frac{1}{x} + x |\tan x|^q, & \text{if } x \neq 0, p, q \in \mathbb{N} \\ 0, & \text{if } x = 0 \end{cases}$

If $f(x)$ is derivable at $x = 0$, then the ordered pair (p, q) is

(1) $(1, 2)$

(2) $(2, 1)$

(3) $(3, 2)$

(4) $(2, 2)$

Q21 - Multiple Correct

Given $f(x) = \begin{cases} b([x]^2 + [x]) + 1, & \text{for } x \geq -1 \\ \sin(\pi(x + a)), & \text{for } x < -1 \end{cases}$ where $[x]$ denotes greatest integer function of x , then for

what values of a and b , the function is continuous at $x = -1$?

(1) $a = 2n + 3/2, b \in \mathbb{R}; n \in \mathbb{I}$

(2) $a = 4n + 2, b \in \mathbb{R}; n \in \mathbb{I}$

(3) $a = 4n + 3/2; b \in \mathbb{R}^+; n \in \mathbb{I}$

(4) $a = 4n + 1, b \in \mathbb{R}^+; n \in \mathbb{I}$

Q22 - Multiple Correct

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If $f(x) = |x + 1|(|x| + |x - 1|)$, then at what points the function is/are not differentiable at in the interval $[-2, 2]$?

- (1) -1
- (2) 0
- (3) 1
- (4) $1/2$

Q23 - Multiple Correct

If $f(x) = [x \sin \pi x]$, then f is (where, $[\cdot]$ denotes greatest integer function)

- (1) continuous at $x = 0$
- (2) continuous in $(-1, 0)$
- (3) differentiable in $(-1, 1)$
- (4) All of these

Q24 - Multiple Correct

Let $f(x) = \cos x$ and $H(x) = \begin{cases} \min[f(t) : 0 \leq t \leq x], & \text{for } 0 \leq x \leq \pi/2 \\ \pi/2 - x, & \text{for } \pi/2 < x \leq 3 \end{cases}$, then

- (1) $H(x)$ is continuous and derivable in $[0, 3]$
- (2) $H(x)$ is continuous but not derivable at $x = \pi/2$
- (3) $H(x)$ is neither continuous nor derivable at $x = \pi/2$
- (4) maximum value of $H(x)$ in $[0, 3]$ is 1

Q25 - Multiple Correct

Let $f : R \rightarrow R$ be a function such that $f(x + y) = f(x) + f(y), \forall x, y \in R$. If $f(x)$ is differentiable at $x = 0$, then

- (1) $f(x)$ is differentiable only in a finite interval containing zero
- (2) $f(x)$ is continuous, $\forall x \in R$

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(3) $f'(x)$ is constant, $\forall x \in R$ (4) $f(x)$ is differentiable except at finitely many points

Q26 - Multiple Correct

 $f(x)$ is continuous at $x = 0$, then which of the following are always true?(1) $\lim_{x \rightarrow 0} f(x) = 0$ (2) $f(x)$ is continuous at $x = 1$ (3) $g(x) = x^2 f(x)$ is continuous at $x = 0$ (4) $\lim_{x \rightarrow 0^+} [f(x) - f(0)] = 0$

Q27 - Multiple Correct

Indicate all correct alternatives, if $f(x) = \frac{x}{2} - 1$, then on the interval $[0, \pi]$ (1) $\tan[f(x)]$ and $\frac{1}{f(x)}$ are both continuous(2) $\tan[f(x)]$ and $\frac{1}{f(x)}$ are both discontinuous(3) $\tan[f(x)]$ and $f^{-1}(x)$ are both continuous(4) $\tan[f(x)]$ is continuous but $\frac{1}{f(x)}$ is not continuous

Q28 - Paragraph 1

Passage I (For Question 28, 29)Consider the function $f(x)$ twice differentiable of degree four symmetrical to line $x = 1$ defined as $f: R \rightarrow R$ and $f''(2) = 0$ The sum of roots of the cubic $f'(x) = 0$ is

(1) 0

(2) 1

(3) 2

(4) 3

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Q29 - Paragraph 1

If $f(1) = 0$ and $f(2) = 1$, then the value of $f(3)$ is

- (1) $6/5$
- (2) $7/5$
- (3) $8/5$
- (4) $9/5$

Q30 - Paragraph 2

Passage II (For Question 30, 31)

Let $f : R \rightarrow R$ be a function defined as $f(x) = \begin{cases} 1 - |x|, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$ and $g(x) = f(x - 1) + f(x + 1), \forall x \in R$

The number of solutions of $f(x) = g(x), |x| < 2$, is/are

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Q31 - Paragraph 2

The number of points at which $g(x)$ is discontinuous, is/are

- (1) 0
- (2) 1
- (3) 2
- (4) 4

Q32 - Paragraph 3

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Passage III (For Question 32, 33)

f be a polynomial function such that $f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ for all $x, y \in [0, \infty)$ and $f(x)$ is one-one function on $[0, \infty)$ with $f(0) = 1$ and $f(1) = 2$.

$f(x)$ is given by

(1) $1 + \frac{2x^3}{3}$

(2) $x^2 + 1$

(3) $1 + \frac{3}{2}x^2$

(4) $1 + \frac{3x^4}{4}$

Q33 - Paragraph 3

The number of discontinuity of $g(x) = \left[\frac{1}{f(x)} \right]$ on $[0, 10]$ is (where, $[.]$ denotes greatest integer function)

(1) 1

(2) 0

(3) 10

(4) 11

Q34 - Paragraph 5**Passage V (For Question 36, 37)**

Let $f(x) = \max \{ |x^2 - 1|, |x - 1| \}$ and $g(x) = \int_a^x f(t) dt, x \in R$

The value of $f(x)$ is

(1) $f(x) = \begin{cases} x^2 - 1, & x \leq -2 \\ 1 - x, & -2 < x \leq 0 \\ 1 - x^2, & 0 < x \leq 1 \\ x^2 - 1, & x > 1 \end{cases}$

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$$(2) f(x) = \begin{cases} x^2 - 1, & x \leq -2 \\ 1 - x^2, & -2 < x \leq 0 \\ 1 - x, & 0 < x \leq 1 \\ x - 1, & x > 1 \end{cases}$$

$$(3) f(x) = \begin{cases} x^2 - 1, & x \leq 1 \\ x - 1, & x > 1 \end{cases}$$

$$(4) f(x) = \begin{cases} x - 1, & x \leq 1 \\ x^2 - 1, & x > 1 \end{cases}$$

Q35 - Paragraph 5

The function $f(x)$ is continuous for x belongs to

$$(1) R - \{0, 1\}$$

$$(2) R - \{-2, 0, 1\}$$

$$(3) R$$

$$(4) \text{None of these}$$

Q36 - Paragraph 6

Passage VI (For Question 38, 39)

Let $f(x) = \begin{cases} [x], & -2 \leq x \leq -1/2 \\ 2x^2 - 1, & -1/2 < x \leq 2 \end{cases}$ and $g(x) = f(|x|) + |f(x)|$, where $[\cdot]$ represents greatest integer function.

The number of points, where $|f(x)|$ is non-differentiable, is

$$(1) 3$$

$$(2) 4$$

$$(3) 2$$

$$(4) 5$$

Q37 - Paragraph 6

The number of points, where $g(x)$ is non-differentiable, is

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(1) 4

(2) 5

(3) 2

(4) 3

Q38 - Paragraph 7

Passage VII (For Question 40, 41)

Let $f(x)$ be a function continuous, $\forall x \in R$ except at $x = 0$. Such that $f'(x) < 0, \forall x \in (-\infty, 0)$ and $f'(x) > 0, \forall x \in (0, \infty)$. Let $\lim_{x \rightarrow 0^+} f(x) = 2, \lim_{x \rightarrow 0^-} f(x) = 3$ and $f(0) = 4$.

The value of λ for which $2 [\lim_{x \rightarrow 0} f(x^3 - x^2)] = \lambda [\lim_{x \rightarrow 0} f(2x^4 - x^5)]$ is

(1) $\frac{4}{3}$

(2) 2

(3) 3

(4) 4

Q39 - Paragraph 7

The value of $\lim_{x \rightarrow 0^+} \frac{f(-x) \cdot x^2}{\left\{ \frac{1 - \cos x}{[f(x)]} \right\}}$, where $[\cdot]$ denotes greatest integer function and $\{\cdot\}$ denotes fractional part function, is

(1) 6

(2) 12

(3) 18

(4) 24

Q40 - Integer Type

Let $a_1, a_2, a_3, \dots, a_n$ be sequence of real numbers with $a_{n+1} = a_n + \sqrt{1 + a_n^2}$ and $a_0 = 0$, if $\lim_{n \rightarrow \infty} \frac{a_n}{2^{n-1}}$ is $\frac{k}{\pi}$, then k is

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Q41 - Integer Type

Suppose $x_1 = \tan^{-1} 2 > x_2 > x_3 > \dots$ are positive real numbers satisfying $\sin(x_{n+1} - x_n) + 2^{-(n+1)} \cdot \sin x_n \cdot \sin x_{n+1} = 0$ for $n > 1$ and $I = \lim_{n \rightarrow \infty} x_n$. Then, the value of $[4I]$ (where, $[\cdot]$ denotes the greatest integer function), is

Q42 - Integer Type

If $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{k^r}{1 \times 3 \times 5 \times 7 \times \dots \times (2r+1)} = 1$, then k^2 is equal to

Q43 - Integer Type

If $a_1 = 1$ and $a_n = n(1 + a_{n-1})$, $\forall n \geq 2$, then $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{a_1}\right) \left(1 + \frac{1}{a_2}\right) \dots \left(1 + \frac{1}{a_n}\right) \right] \cdot \frac{1}{e}$

Q44 - Integer Type

Let $f(x) = [x] + [x + 1/4] + [x + 2/4] + [x + 3/4]$. Then, the number of points of discontinuity of $f(x)$ in $[0, 1]$ is (where, $[\cdot]$ denotes greatest integer function)

Q45 - Integer Type

If $f(x)$ is a continuous function, $\forall x \in \mathbb{R}$ and the range of $f(x)$ is $(2, \sqrt{21})$ and $g(x) = \left[\frac{f(x)}{c} \right]$ is continuous, $\forall x \in \mathbb{R}$, then the least positive integral value of c is (where, $[\cdot]$ denotes greatest integer function)

Q46 - Integer Type

The only integer lying in the interval $(0, 4)$, where the function $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{\pi x}{2} \right)^{2n}$ is discontinuous, is

Q47 - Integer Type

Let $f(x) = x^3 - x^2 + x + 1$, $g(x) = \begin{cases} \max[f(t)], 0 \leq t \leq x, & \text{if } 0 \leq x \leq 1 \\ 3 - x, & \text{if } 1 < x \leq 2 \end{cases}$, then the number of points in $(0, 2)$, where $g(x)$ is not differentiable is

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Q48 - Integer Type

The number of points of discontinuity of $f(x) = [2x]^2 - \{2x\}^2$ (where, $[\cdot]$ and $\{\cdot\}$ denote greatest integer and fractional part respectively) in the interval $(-2, 2)$, are

Q49 - Integer Type

Let $f(x)$ be a function which is differentiable any number of times $f(2x^2 - 1) = 2x^3 f(x)$, $\forall x \in R$, then $f^{2016}(0)$ is equal to (where, $f^n(x) = n$ th order derivative of f at x)

Q50 - Integer Type

The number of points, where $f(x) = [\sin x + \cos x]$ (where, $[\cdot]$ denotes greatest integer), $x \in (0, 2\pi)$ is not continuous, is

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Answer Key

Q1 (1)

Q2 (1)

Q3 (1)

Q4 (4)

Q5 (2)

Q6 (2)

Q7 (2)

Q8 (4)

Q9 (1)

Q10 (2)

Q11 (4)

Q12 (2)

Q13 (3)

Q14 (2)

Q15 (2)

Q16 (1, 3)

Q17 (1, 2, 3, 4)

Q18 (2, 3, 4)

Q19 (1, 4)

Q20 (2, 4)

Q21 (1)

Q22 (1, 2, 3)

Q23 (4)

Q24 (1, 4)

Q25 (2, 3)

Q26 (3, 4)

Q27 (3, 4)

Q28 (4)

Q29 (3)

Q30 (2)

Q31 (1)

Q32 (2)

Q33 (1)

Q34 (1)

Q35 (3)

Q36 (1)

Q37 (4)

Q38 (3)

Q39 (2)

Q40 (4)

Q41 (3)

Q42 (4)

Q43 (1)

Q44 (4)

Q45 (5)

Q46 (2)

Q47 (1)

Q48 (6)

Q49 (0)

Q50 (5)

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